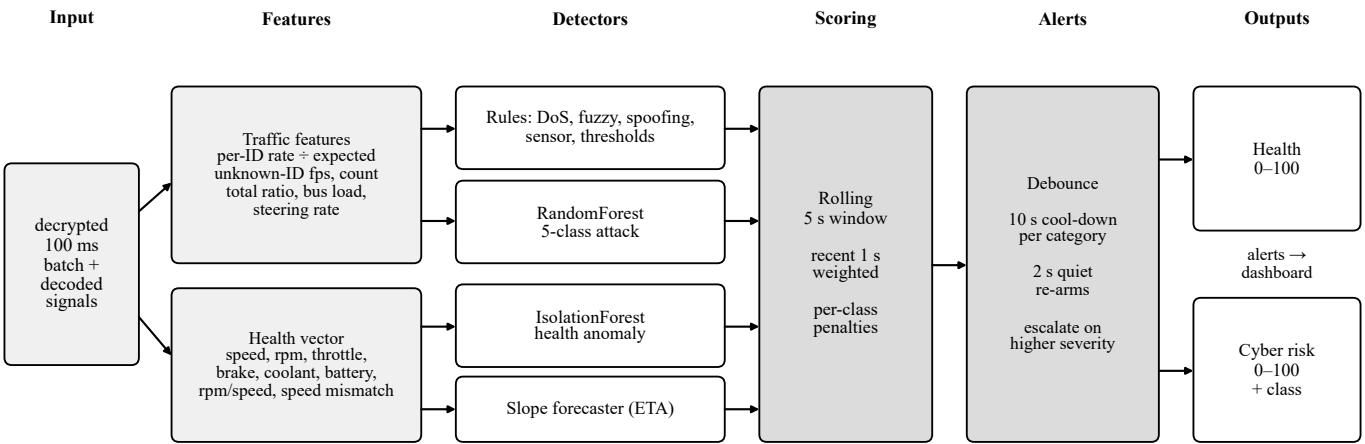

Input
Features
Detectors
Scoring
Alerts
Outputs
decrypted 100 ms batch + decoded signals
Traffic features per-ID rate ÷ expected unknown-ID fps, count total ratio, bus load, steering rate
Health vector speed, rpm, throttle, brake, coolant, battery, rpm/speed, speed mismatch
Rules: DoS, fuzzy, spoofing, sensor, thresholds
RandomForest 5-class attack
IsolationForest health anomaly
Slope forecaster (ETA)
Rolling 5 s window
recent 1 s weighted
per-class penalties
Debounce
10 s cool-down per category
2 s quiet re-arms
escalate on higher severity
Health 0–100
alerts → dashboard
Cyber risk 0–100 + class
Executed once per received batch; per-vehicle state (one analyzer per device_id)